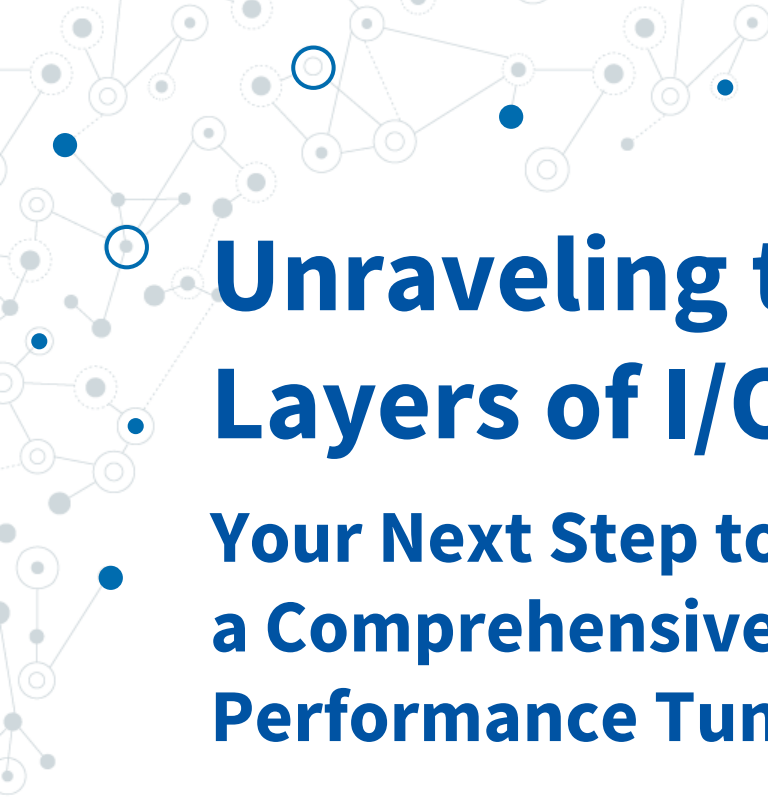


A decorative graphic in the top-left corner consisting of a network of interconnected nodes and lines, with some nodes highlighted in blue and others in grey.

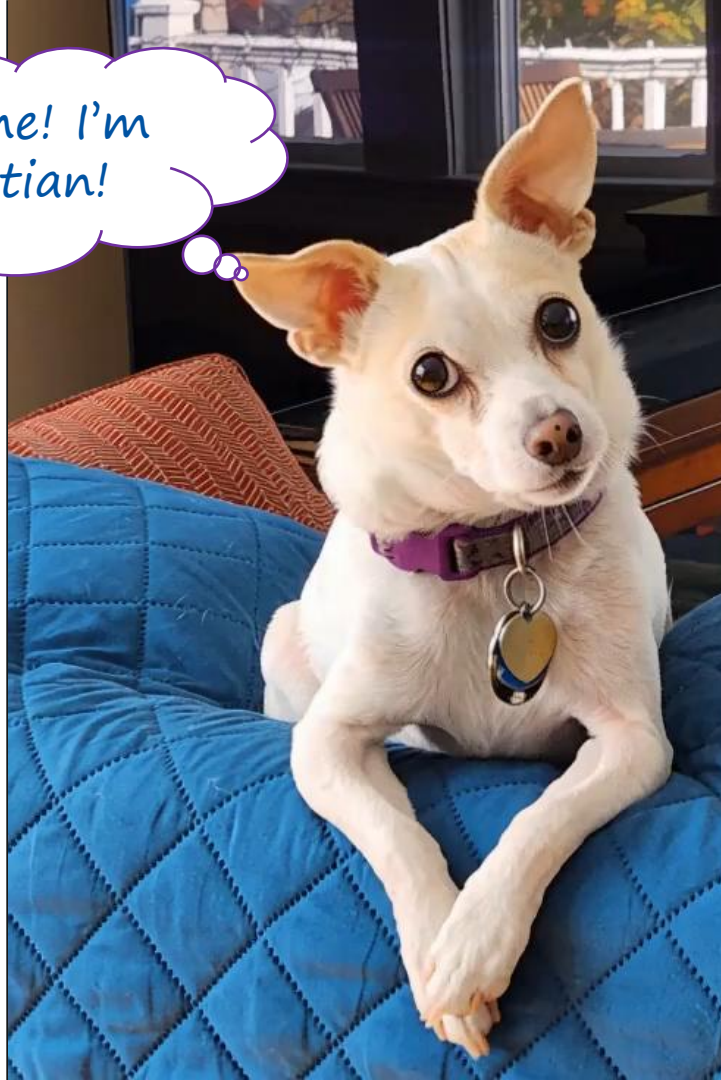
Unraveling the Layers of I/O:

**Your Next Step to Becoming
a Comprehensive
Performance Tuner**

Andy Yun

sqlbek@gmail.com

*Welcome! I'm
Sebastian!*





Blegh...

Yun

ation Architect

- SQL Server DBA & DB Developer
- Server Administrator
- SQL Server 2008 R2



Group
Director-at-Large
Organizer



sqlbek@gmail.com

<https://sqlbek.wordpress.com/>

<https://www.github.com/sqlbek/>



I got paged FOURTEEN TIMES over the weekend!!

This app database has SO MANY performance problems! I/O just SUCKS!!!

Wait, you have a job...?!?



More importantly, HOW is the dog talking to us via Slack?



Forget that... since you cut back on my biscuit allowance, I had to make ends meet!

I've been moonlighting for StinkyWinds as their on-call DBA for 3 months

?!?... How'd... you even pass an interview?!?



Look, I've been listening to you & Deb babble about SQL for years... I've learned a thing or three about T-SQL performance tuning.

I know enough to be dangerous!



See, StinkyWinds's problems are more than just security & terrible T-SQL

There's issues with disk... network... I/O... I'm so confused!!! How does it all work!?!?



And someone keeps trying to blame fragmentation over and over! **HOWLS & CRIES**

Sebastian, sit... SIT... **SIT!**

Okay, thank you! Okay, let's unpack all of this



Learning Journey

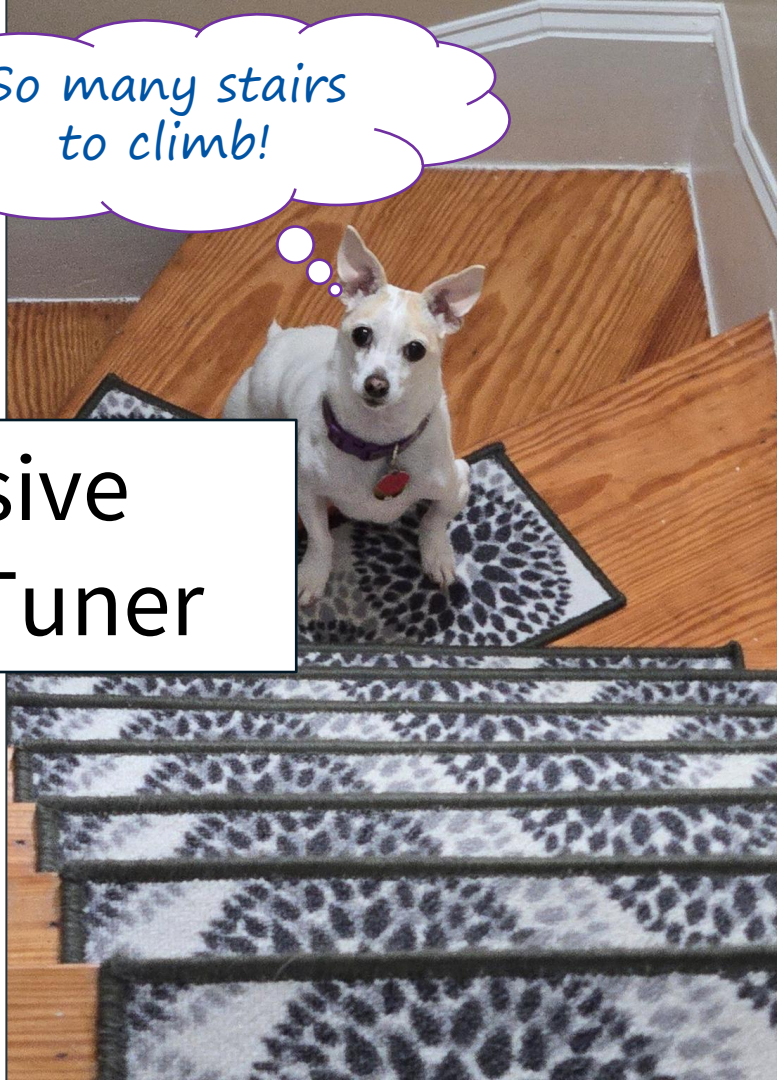
◎ T-SQL Developer

◎ T-SQL

◎ SQL Server Performance Tuner

Comprehensive
Performance Tuner

*So many stairs
to climb!*



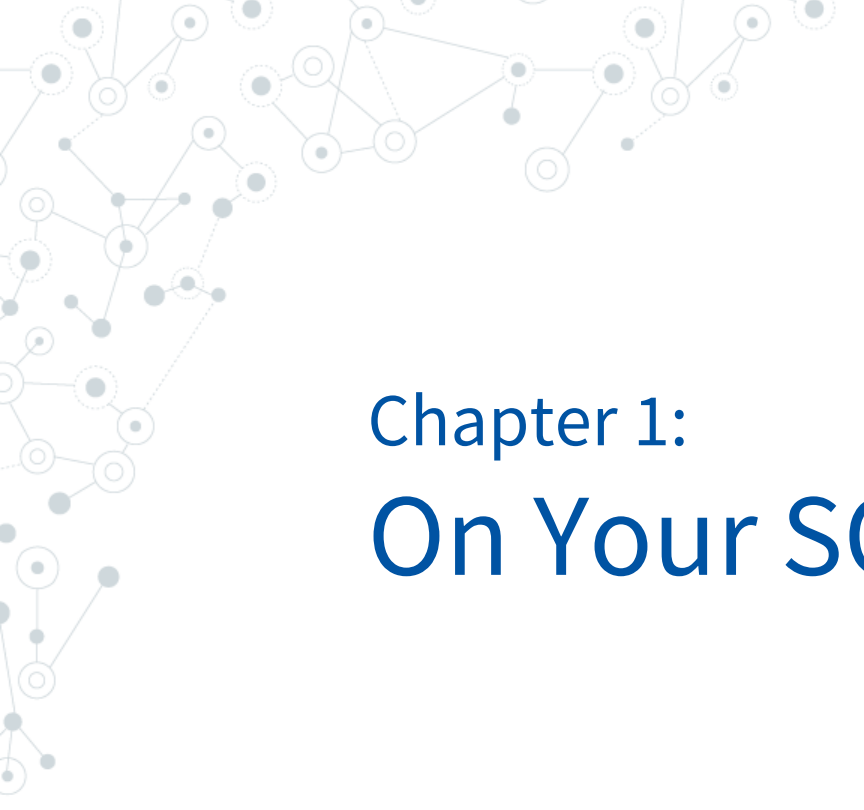
Agenda

Progression of I/O

- ◎ On Your SQL Server
- ◎ Server to Storage
- ◎ Beyond the Storage Fabric
- ◎ *Part 4 of a Series*
- ◎ Advanced for EightKB

*I'm fast
too!*



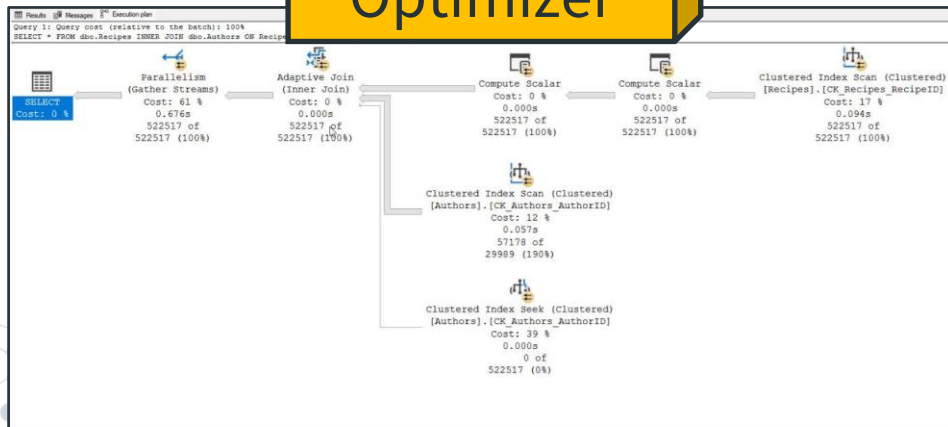
A decorative network diagram in the top-left corner, featuring a complex web of interconnected nodes and lines. The nodes are represented by small circles, some of which are larger and have concentric circles, suggesting a hierarchical or central structure. The lines are thin and gray, connecting the nodes in a non-linear fashion.

Chapter 1: On Your SQL Server

```
SELECT *  
FROM dbo.Recipes  
INNER JOIN dbo.Authors  
ON Recipes.AuthorID = Authors.AuthorID
```

- ◎ Tell me WHAT you want...
- ◎ I'll decide HOW...

Query
Optimizer



Yes, he
cooks

Get Me a Data Page

Physical Read:

Read a Data Page
from Disk into the Buffer Pool

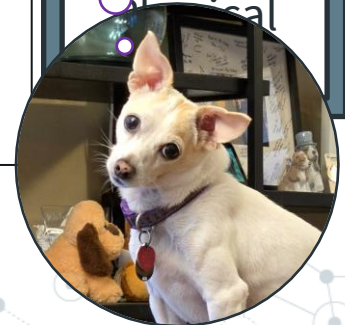
Logical Read:

Scan or “Processing” of a Data Page
inside the Buffer Pool

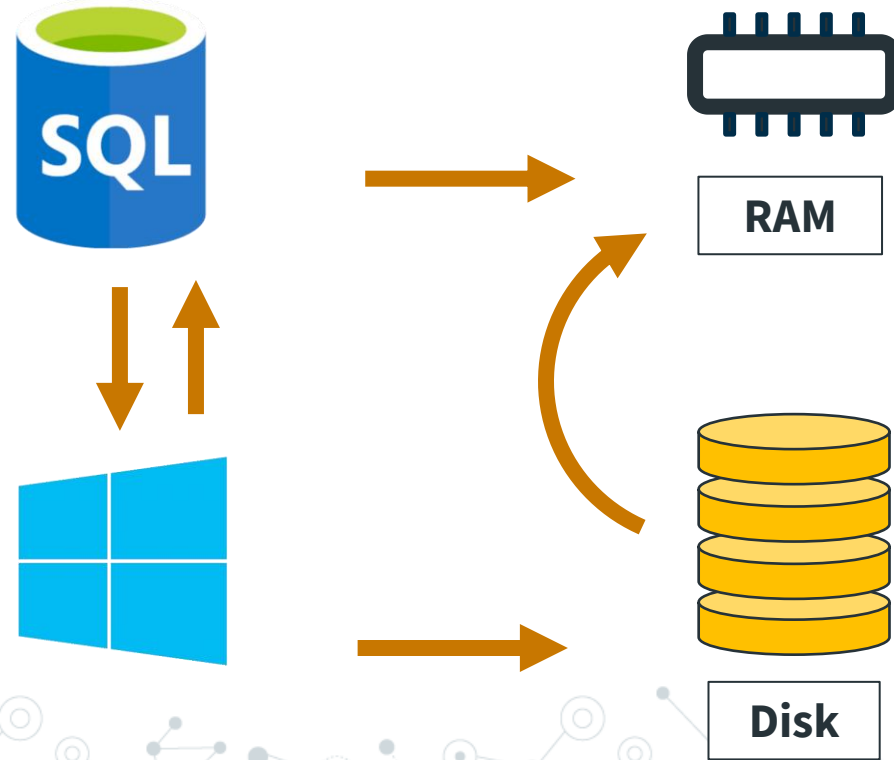
Is a Physical
Read an
“IOP?”

Read the
data page

Logical
Read



Who Dispatches I/O Requests?



“I Need a Data Page from Disk” Lifecycle

(SQL) Worker:

1. Reserve an empty 8KB frame in the buffer pool

- ◎ **PAGEIOLATCH_EX**

2. Call Windows API: ReadFileScatter

3. Windows says “Okay, I’m on it”

4. SQL Worker goes and does something else

- ◎ Asynchronous

- ◎ More I/O in-flight = greater concurrency

Scatter/Gather I/O

ReadFileScatter / WriteFileGather

- ◎ Read: Consume contiguous data from disk
- ◎ Scatter into different separate memory buffers
- ◎ Write: Consume from different memory buffers
- ◎ Gather into a continuous write stream to disk

An Added Twist

(SQL) Worker:

1. I've issued all my Requests for Data Pages
2. But I still have to read them!
3. Check the buffer pool – not there

© **PAGEIOLATCH_SH**

“Waiting on yourself” – Bob Dorr

♪ Dancing
with
myself! ♪



IRP – I/O Request Packet

- ⦿ Window's "container" for I/O
- ⦿ Built by Windows I/O manager
- ⦿ Disk class driver builds the SCSI commands
- ⦿ Storage port driver may split the IRP
- ⦿ Output -> "a Storage IOP"

- 
1. File System driver (NTFS)
 2. Volume mgr.
 3. Partition mgr.
 4. Disk class driver
 5. Storage port driver
 6. I/O Dispatched

Filter Drivers

◎ File System Minifilters

- Antivirus, backup agents, CrowdStrike & Windows Defender

◎ Volume Filters

- BitLocker, Changed Block Trackers (CBT) – ex: Veeam

◎ Device / Bus Filters

- MPIO, Block Replication

1. File System driver (NTFS)
2. Volume mgr.
3. Partition mgr.
4. Disk class driver
5. Storage port driver
6. I/O Dispatched

Chapter 1.2:

Unraveling Fragmentation

*Is that still
a thing?!*



Fragmentation (*“External”*)

Definition:

- ⦿ In B-tree (rowstore) indexes, fragmentation exists when indexes have pages in which
 - ⦿ ... the logical ordering within the index
 - ⦿ ... based on the key values of the index
 - ⦿ ...doesn't match the physical ordering of index pages.

*“...physical ordering
of index pages...”*

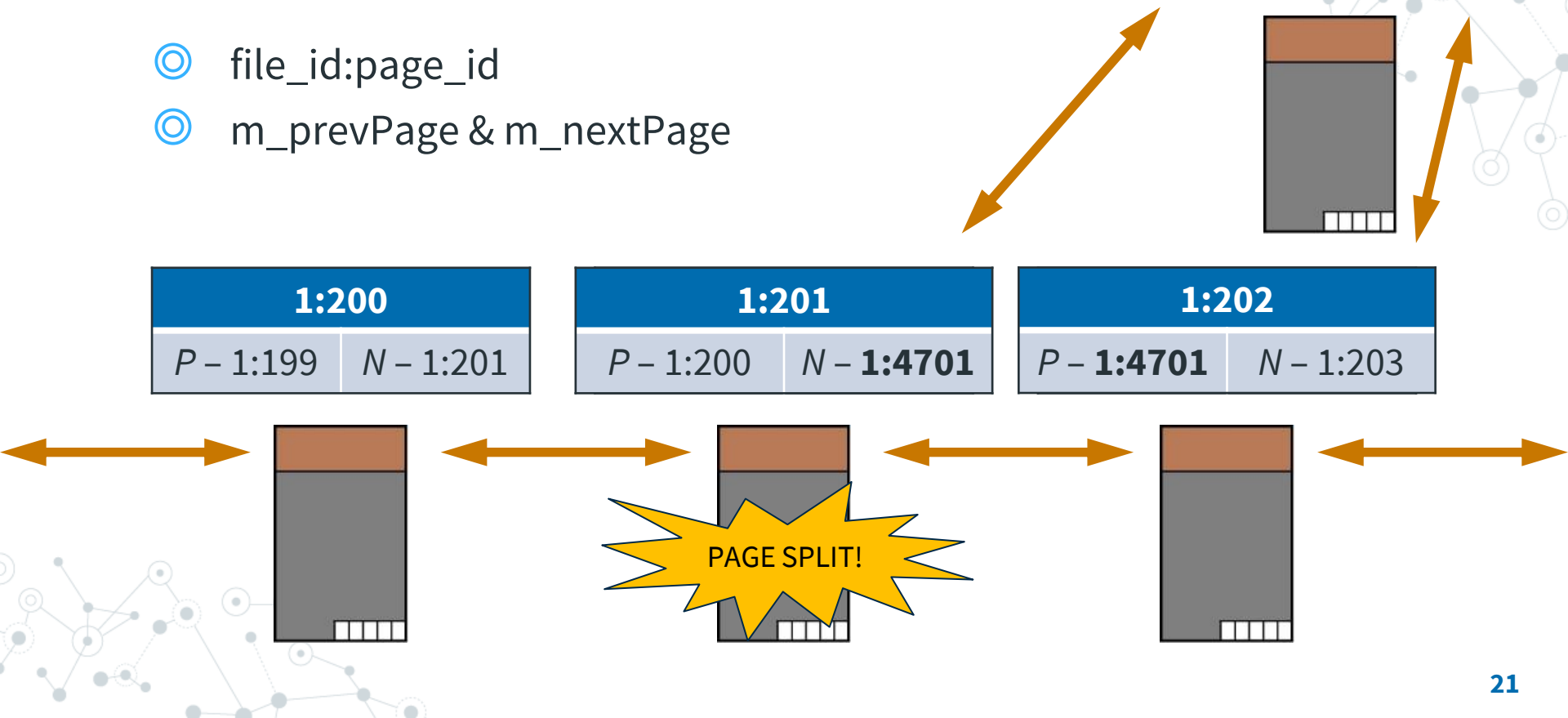
On Actual Disk?

- ⦿ Yes?
- ⦿ No?

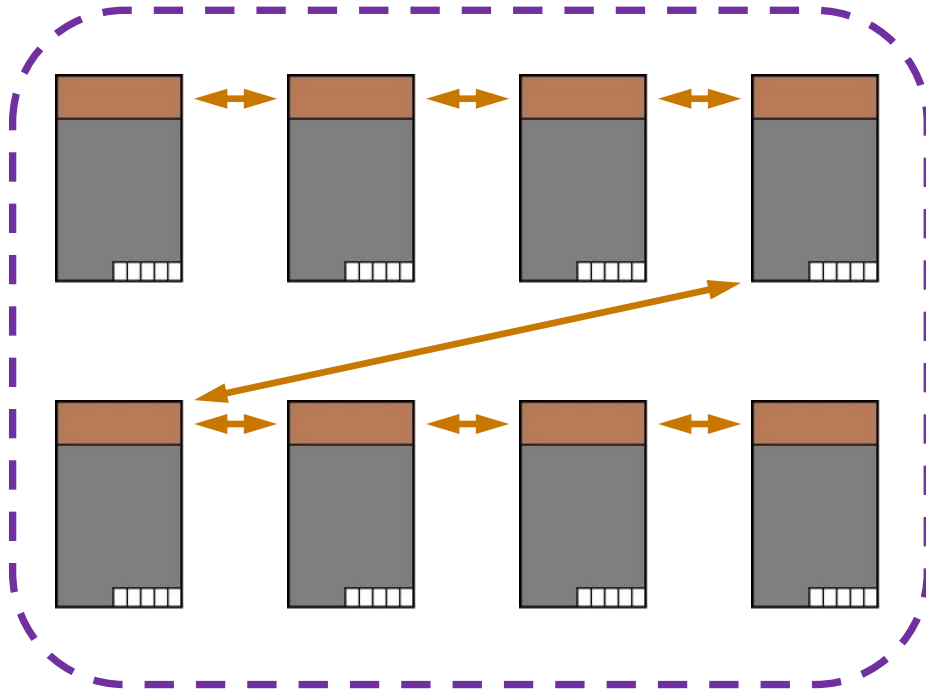


Data Pages & Linked Lists

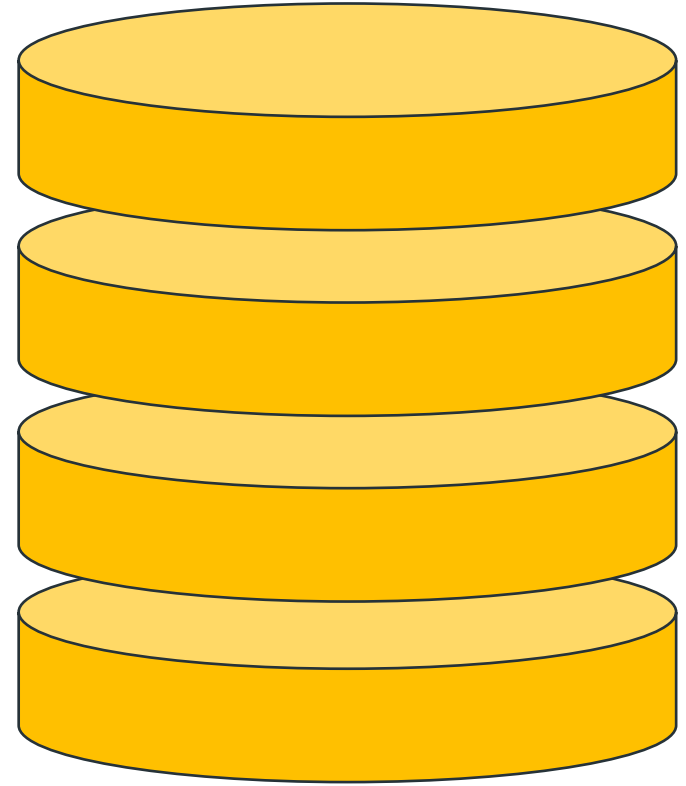
- ⦿ file_id:page_id
- ⦿ m_prevPage & m_nextPage



Layers and Layers...



MDF/NDF
Data File



Physical
Storage

“Physical” Deciphered

- ◎ Page IDs are File Offsets in the MDF
 - Page ID * 8192 = file offset
 - 1:250 -> 250 * 8192 -> 2,048,000 bytes
- ◎ **Physical location** of that data page INSIDE the MDF/NDF!
- ◎ A MDF/NDF is a byte stream on top of File System
 - ◎ Contiguous on disk?
 - ◎ File System fragmented?
 - ◎ Physically abstracted entirely?

Readahead Reads

- ◎ Typical read = single 8KB data page
 - Not extents?
 - Extent is an *allocation unit*, not an I/O unit
- ◎ Reads up to 64 contiguous pages (512KB) as a single Scatter/Gather call
- ◎ m_prevPage & m_nextPage *

(There's also "random prefetch" but that's another show...)

* And there's another caveat but hold that thought

Ada	501
Carl	502
Felix	503
Iris	504
Liam	505
Rose	506



501
Ada
Alan
Amy
Ben
Bruno

502
Carl
Clara
Dan
Dylan
Eric

503
Felix
Fiona
Grace
Henry
Ian

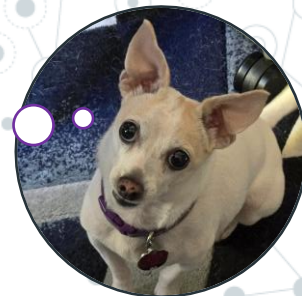
504
Iris
Jack
Jane
Kay
Leo

505
Liam
Maya
Noah
Owen
Paul

506
Rose
Sam
Tara
Vera
Zoe

Ada	501
Carl	1441
Felix	1442
Iris	61
Liam	101
Rose	102

*That'll be four
I/O requests,
two ReadAheads*



501
Ada
Alan
Amy
Ben
Bruno

1441
Carl
Clara
Dan
Dylan
Eric

1442
Felix
Fiona
Grace
Henry
Ian

61
Iris
Jack
Jane
Kay
Leo

101
Liam
Maya
Noah
Owen
Paul

102
Rose
Sam
Tara
Vera
Zoe

Ada	501
Carl	1441
Felix	242
Iris	61
Liam	9201
Rose	102

*Can't ReadAhead
– that's totally
fragmented!*

*Didn't you say
there was a
caveat?*



501
Ada
Alan
Amy
Ben
Bruno

1441
Carl
Clara
Dan
Dylan
Eric

242
Felix
Fiona
Grace
Henry
Ian

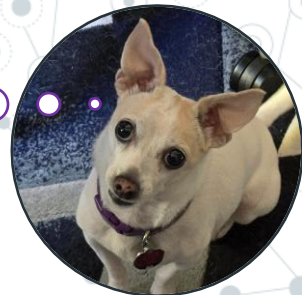
61
Iris
Jack
Jane
Kay
Leo

9201
Liam
Maya
Noah
Owen
Paul

102
Rose
Sam
Tara
Vera
Zoe

Ada	100
Carl	99
Felix	98
Iris	97
Liam	96
Rose	95

*You shrank a
database?!*



- DMV = ~99% fragmentation
- Still got Read Aheads
- Allocation Order Scans**

100
Ada
Alan
Amy
Ben
Bruno

99
Carl
Clara
Dan
Dylan
Eric

98
Felix
Fiona
Grace
Henry
Ian

97
Iris
Jack
Jane
Kay
Leo

96
Liam
Maya
Noah
Owen
Paul

95
Rose
Sam
Tara
Vera
Zoe

Process Monitor

File Edit Help

Time of Day

4:13:43.21461...
4:13:43.21575...
4:13:43.21622...
4:13:43.21655...
4:13:43.21673...
4:13:43.21689...
4:13:43.21708...
4:13:43.21733...
4:13:43.21745...
4:13:43.21756...
4:13:43.21761...
4:13:43.21776...
4:13:43.21787...
4:13:43.21801...
4:13:43.21814...
4:13:43.21826...
4:13:43.21842...
4:13:43.21860...
4:13:43.21873...
4:13:43.21889...

Detail

Offset: 6,160,384, Length: 65,536, I/O Flag
Offset: 11,599,872, Length: 65,536, I/O Flag
Offset: 7,331,840, Length: 524,288, I/O Flag
Offset: 7,856,128, Length: 524,288, I/O Flag
Offset: 8,380,416, Length: 524,288, I/O Flag
Offset: 8,904,704, Length: 524,288, I/O Flag
Offset: 9,428,992, Length: 8,192, I/O Flag
Offset: 9,502,720, Length: 65,536, I/O Flag
Offset: 9,633,792, Length: 524,288, I/O Flag
Offset: 10,158,080, Length: 524,288, I/O Flag
Offset: 10,682,368, Length: 524,288, I/O Flag
Offset: 11,206,656, Length: 393,216, I/O Flag
Offset: 3,211,264, Length: 524,288, I/O Flag
Offset: 3,735,552, Length: 524,288, I/O Flag
Offset: 4,259,840, Length: 524,288, I/O Flag
Offset: 4,784,128, Length: 524,288, I/O Flag
Offset: 5,308,416, Length: 524,288, I/O Flag
Offset: 5,832,704, Length: 327,680, I/O Flag
Offset: 6,291,456, Length: 524,288, I/O Flag
Offset: 6,815,744, Length: 516,096, I/O Flag

Event Properties

Event Process Stack

Frame	Module	Location	Address
K 0	FLTMGR.SYS	FltDecodeParameters + 0x2616	0xffff80211d...
K 1	FLTMGR.SYS	FltDecodeParameters + 0x20b6	0xffff80211d...
K 2	FLTMGR.SYS	FltDecodeParameters + 0xbad	0xffff80211d...
K 3	FLTMGR.SYS	FltDecodeParameters + 0x56b	0xffff80211d...
K 4	ntoskml.exe	IoCallDriver + 0x55	0xffff8020e2...
K 5	ntoskml.exe	SePrivilegeCheck + 0x2d8e	0xffff8020e6...
K 6	ntoskml.exe	NtReadFileScatter + 0x737	0xffff8020e7...
K 7	ntoskml.exe	setimpex + 0x8b75	0xffff8020e4...
U 8	ntdll.dll	ZwReadFileScatter + 0x14	0x7f97c7a07...
U 9	KERNELBASE.dll	ReadFileScatter + 0x6c	0x7f979d432...
U 10	sqlmin.dll	FCB::ScatterReadInternal + 0x1ba	0x7f95ae8da...
U 11	sqlmin.dll	FCB::ScatterRead + 0x326	0x7f95ae8de...
U 12	sqlmin.dll	FCB::ScatterRead + 0x783	0x7f95ae8e3...
U 13	sqlmin.dll	FCB::ScatterRead + 0xedb	0x7f95ae8ea...
U 14	sqlmin.dll	Page::Init + 0x518	0x7f95ae801...
U 15	sqlmin.dll	DbLedger::GetNextSlot + 0x760	0x7f95af090...
U 16	sqlmin.dll	DbLedger::GetNextSlot + 0xb91	0x7f95af095...
U 17	sqlmin.dll	XmlLongWaitGraphWriter::XmlLongWaitGraphWriter + 0x123a	0x7f95b0269...
U 18	sqlmin.dll	DatasetSession::WakeUp + 0x1893	0x7f95ae53f...
U 19	sqlmin.dll	Page::GetHeader + 0x9a8	0x7f95ae707...
U 20	sqlmin.dll	GetSqlServerGlobals + 0xc07	0x7f95b0491...
U 21	sqlmin.dll	Page::GetHeader + 0xc88	0x7f95ae70a...
U 22	sqlmin.dll	PageKey::PageKey + 0x61f5	0x7f95b047e...
U 23	sqlmin.dll	Page::GetHeader + 0x3411	0x7f95ae731...
U 24	sqlmin.dll	XeSqlPkg::rpc_starting_aggregate_xdb::IsEnabled + 0x7d7	0x7f95aebb5...

What's Queue
Depth?

*Does fragmentation
really matter anymore?*

- ◎ **REALLY** terrible storage?
- ◎ Throttled on IOPS - cloud?
- ◎ Low Queue Depth settings?

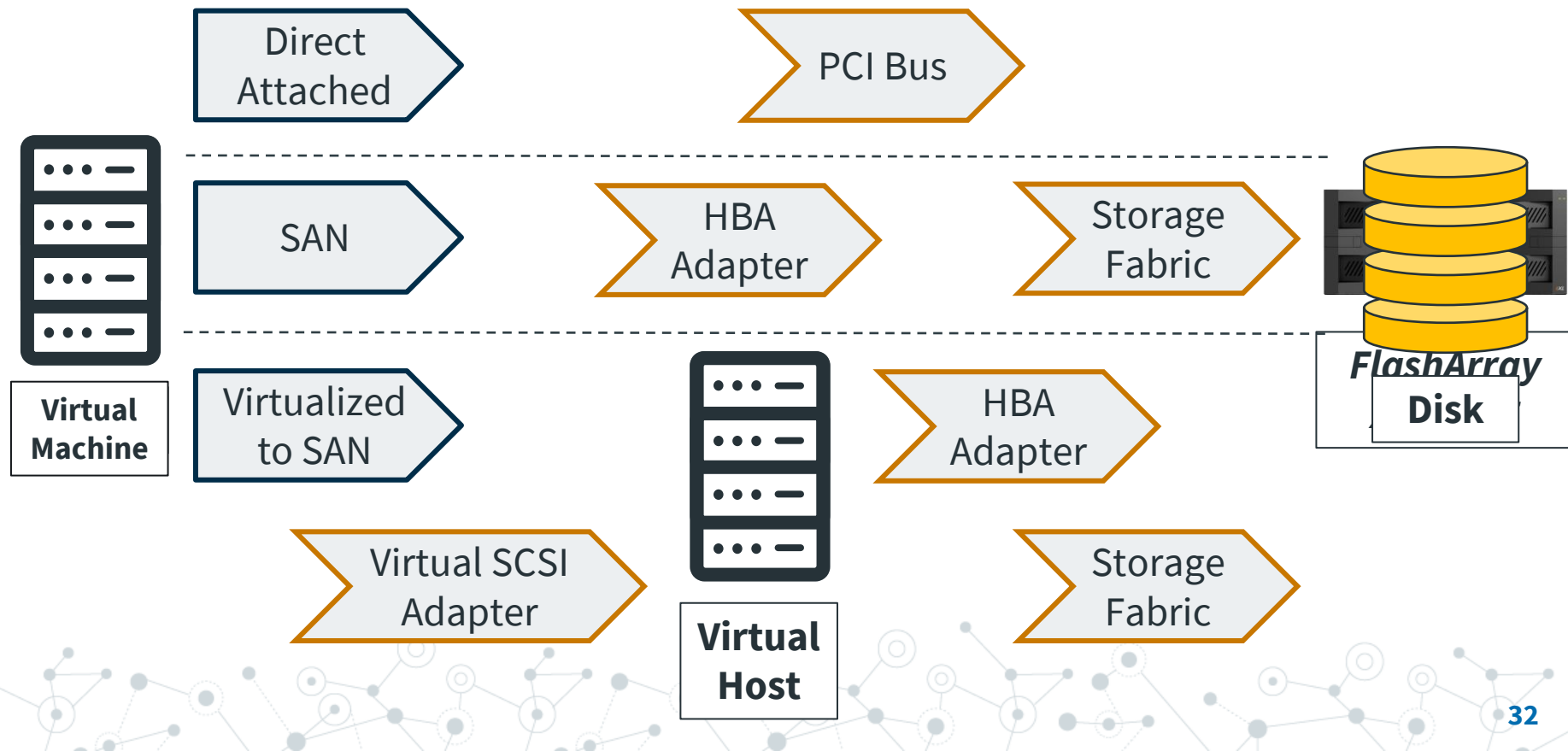


Chapter 2: Server to Storage

*Is that
somewhere
to walk to?
That sounds
like a walk!*



Physical Pathway Refresher



Refresher

- ⦿ Latency – Time for one I/O completion
 - ⦿ aka. A I/O's “round trip”
- ⦿ IOPS – How many complete per second
- ⦿ $\text{Throughput} = \text{IOPS} \times \text{I/O size}$

Remember
when 10ms
was fast?



Storage is Faster Now

- Do we care whether we fire many small I/Os instead of a few readaheads?
- SQL I/O is asynchronous - “fire and forget”
- But can the destination handle a flood of I/O?



Little's Law

$$\text{Throughput} = \frac{\text{Queue Length}}{\text{Latency}}$$

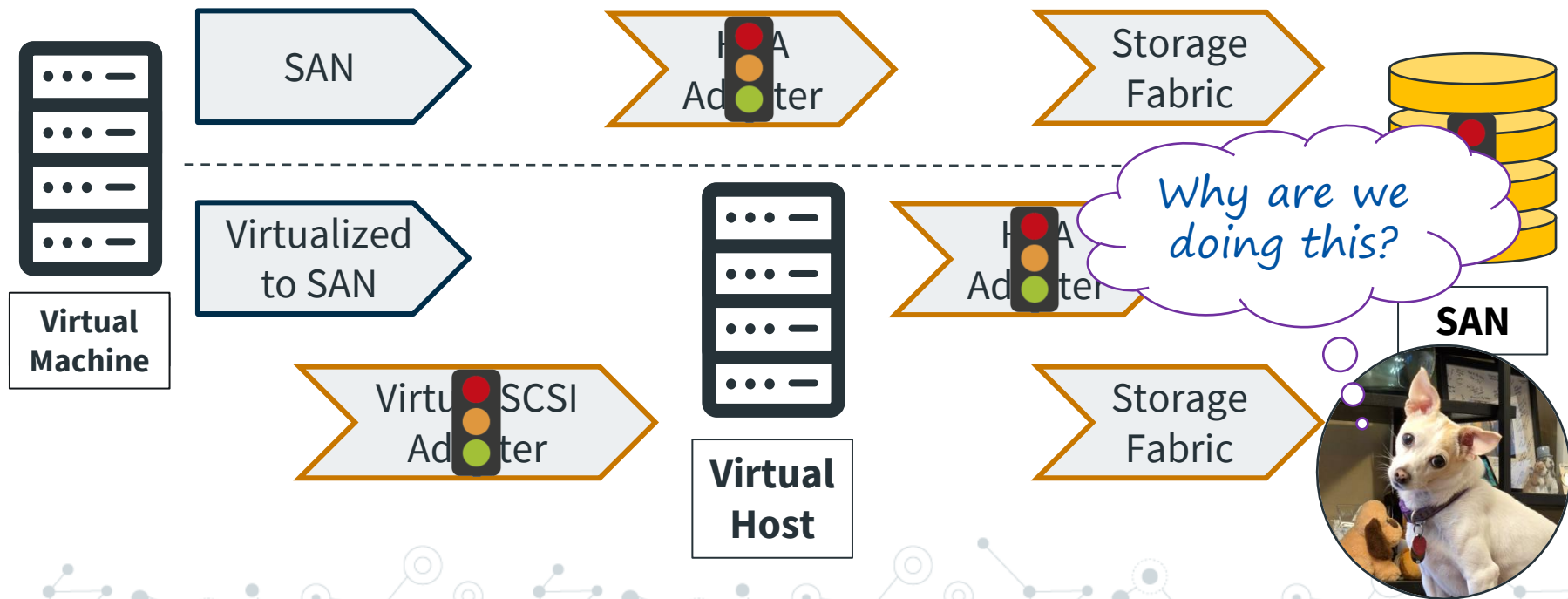
- ❖ **Queue Depth (QD):**

Maximum number of I/O requests that can be outstanding at once

- ❖ **Queue Length (QL):**

Total number of I/O requests currently in-flight

Queues Everywhere





Remember the block party & all the neighborhood kids giving you treats all at once?



Best Day EVAR!!!



And how we give you treats normally?



*One at a time, 'cause y'all are slow!
But the kids... mmm, TREAT HURRICANE!!!
I never stopped chewing.*



Right, not because the kids were any faster. But because there were 10 of them all holding out treats for you SIMULTANEOUSLY!

And that's concurrency! You were never waiting!
And how many kids did we let crowd around you? That's queue depth!



When's the next block party?!?!?

Confusing & Overloaded Vocabulary

- ◎ **Outstanding** = I/Os that are accepted and held
 - But not necessarily being serviced yet
- ◎ Queue Length = may mean “queue occupancy” and/or “queue overflow or wait count”

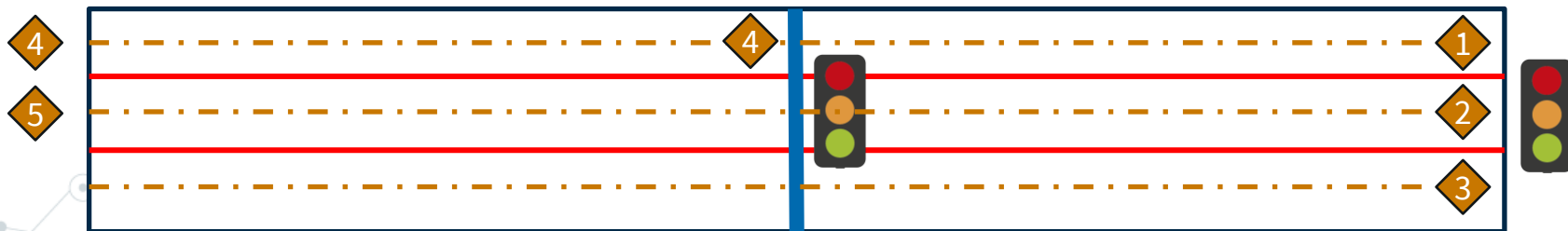
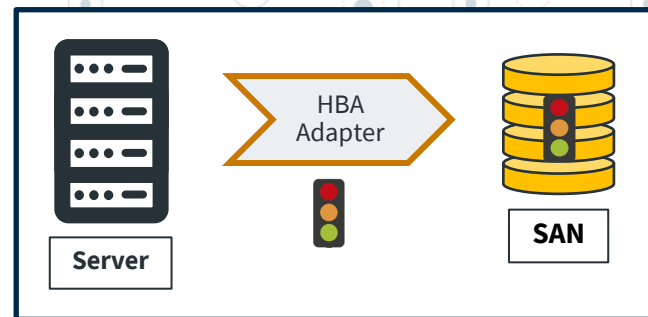
- ◎ **Queue Depth (QD):**
Maximum number of I/O requests that can be **outstanding** at once
- ◎ **Queue Length (QL):**
Total number of I/O requests currently **in-flight**

*This
ALWAYS
confuses
me!*



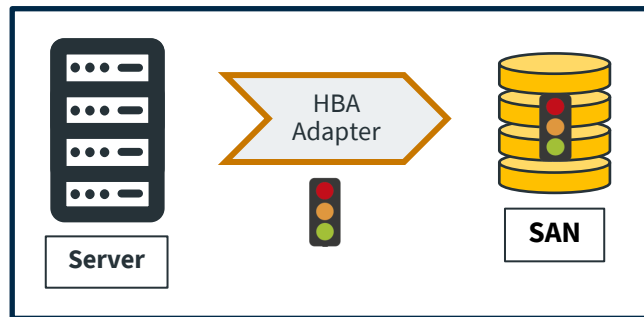
Multiple Queues

- Queue in HBA Adapter & SAN
- Both have Queue Depth of 32
- Does this mean each queue “holds” 32 I/O’s each?
- NO - All queues in the pathway OVERLAP



Math Time

*Server dispatches
200 I/Os*

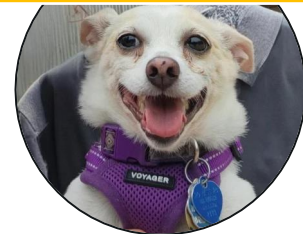


	HBA Queue Depth	SAN Queue Depth	HBA Occupancy	SAN Occupancy	Waiting Inside HBA	Held above (server)
1	128	128	128	128	0	72
2	128	32	128	32	96	72
3	32	128	32	32	0	168

Storage: Then & Now

	Disk SAN (~2010-2015)	All-flash SAN defaults/unoptimized	All-flash SAN optimized
1 million data pages / 7.63 GiB			
Latency per I/O	10 ms	0.5 ms	0.5 ms
Effective Queue Depth	32	32	256
Achievable IOPS	3,200	64,000	512,000
Throughput	~25 MB/s	~512 MB/s	~4GB/s
Perf vs Spinning Disk	1x	20x	160x

Hey, we've
increased
concurrency!



REMEMBER:

Almost every hop in your I/O path involves a queue

- ◎ *The shallowest queue depth becomes the max for the entire path!*

I don't have a queue depth for treats!

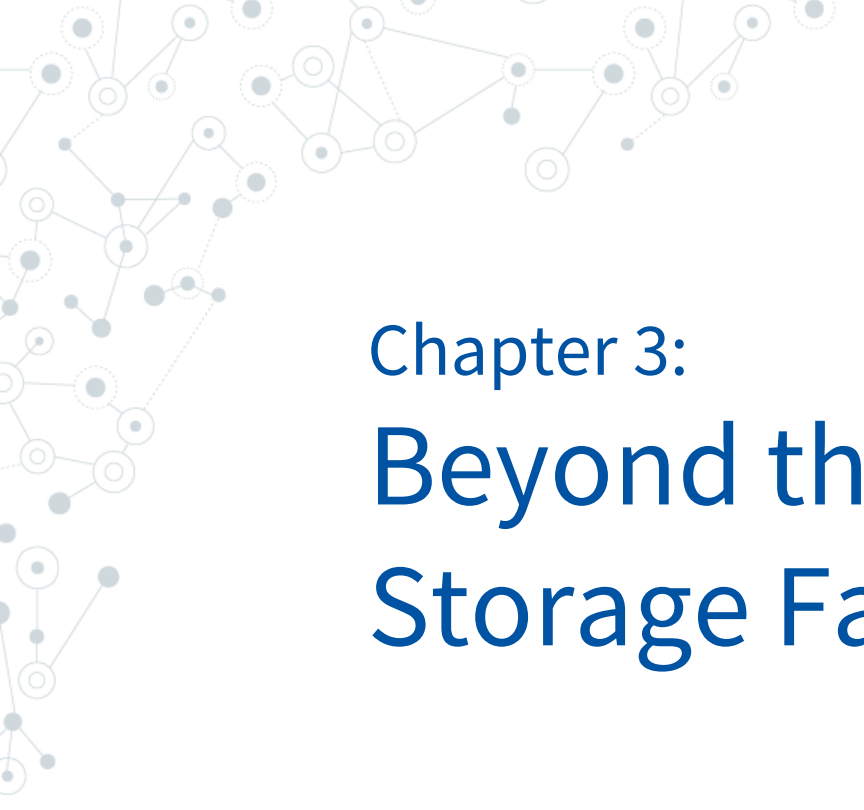


Does fragmentation really matter anymore?

*More important
things, like
napping...*



- ⦿ *Back then – cost seek time*
- ⦿ *Now – more I/O volume,
need larger queues*

A decorative network diagram in the top-left corner, featuring a complex web of interconnected nodes and lines. The nodes are represented by small circles, some of which are larger and have concentric circles, suggesting different levels of connectivity or data flow. The lines are thin and gray, creating a mesh-like structure.

Chapter 3: Beyond the Storage Fabric



So the other thing that's been plaguing me is this pesky Availability Group on 2019, that keeps lagging horribly. Help?



Well tell us more. What's changed recently?



So this StinkyWinds system does a nightly data load. And the amount of data has quadrupled!

Well, there's fundamentally three places that I/O will stall out. We can unpack that if you like?





Here's the other thing that confuses me. They upgraded the network link between the AG's. Latency is okay but bandwidth is only like 30% utilized???

K – sit tight, I think we may be onto something...



AG Log Block Journey: The Pipeline

LOG SEND QUEUE



Primary
Replica

B - Ship across the wire

STALL 1?

A + B run in parallel

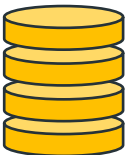
D - Acknowledge



Secondary
Replica

A - Harden locally

WRITELOG



SAN 1

E - Commit
returns to the
app

C - Harden on the secondary

WRITELOG (secondary replica)

HADR_SYNC_COMMIT

REDO QUEUE

STALL 3?

Redo: apply log to data files

Secondary data
files

STALL 2?



SAN 2

3 Stall Points

- ⦿ Stall 1 – Sending log blocks to replica(s)
- ⦿ Stall 2 – Hardening log entries on a replica
- ⦿ Stall 3 – Redo queue applying logs to data files
- ⦿ 2 & 3 – Disk I/O related
- ⦿ 1 – Network related
 - Flow Control & UCS Boxcars

Flow Control

⦿ Flow Control Gate

- 1 log block = 1 message
- Limits # of simultaneous outstanding messages
- Two limit values: per database AND transport total

⦿ Perfmon:

Flow control/sec and Flow Control Time (ms/sec);

⦿ XE: database_transport_flow_control_action hadron_database_flow_control_action

UCS Boxcars

- ◎ Universal Communication Service - protocol
- ◎ Groups packets of UCS data into a “container” for shipping
- ◎ max ucs send boxcars – limits # of boxcars outstanding

Log Block to Network Packet

- ◎ Log block – 512bytes -> 60KB = 1 message
 - Flow Control gates # of messages in-flight
- ◎ 1 message -> serialized into 1 or more UCS packets
- ◎ UCS packets grouped into a UCS boxcar
 - max ucs send boxcars gates # of box cars in-flight
- ◎ UCS boxcar written to a TCP connection
 - TCP chops UCS boxcar into TCP network packets
- ◎ **Latency is the enemy of everything here**

Improvements in SQL Server 2022

⦿ Flow Control Gate increased

- Total limit: 8,192 -> 16,384
- Per database: 1,792 -> 7,168

⦿ Default in 2022

- Backported to 2019 CU9, 2017 CU18, and 2016 SP1 CU16 behind TF 12310
- TF is a start-up parameter only!

Improvements in SQL Server 2025

⦿ Max ucs send boxcars increased

- Default: 256
- Tunable up to 2048 now

⦿ sp_configure in 2025

- Registry value in 2022



So you said the amount of data quadrupled. Why not try the Flow Control adjustment first. Check the Flow Control related XE events and Perfmon counters.



While you were babbling, I already flipped the Trace Flag 12310 and wow, we're actually saturating our link now!

Wait, you said that was a 2019 instance. You restarted a prod server during the day?!



Oh don't worry, everyone at StinkyWinds takes 3 hour lunches. Easy daily maintenance windows!

Conclusion

*Is it almost
time for my
walk?!*



Recap

- ◎ I/O isn't one layer, it's many.
- ◎ Queues... are everywhere...
- ◎ Practically every default in your stack was set for hardware that no longer exists.
- ◎ The bottleneck didn't disappear – it just moved.

This is the Trilogy's Epilogue

Let's Dive Into SQL Server I/O To Improve
T-SQL Performance

<https://youtu.be/fDd4lw6DfqU>

Is Storage the Root Cause of Your
Performance Woes... or Not?

<https://youtu.be/kL9UVME6PO4>

Practical Deep Dive into IO for T-SQL
Performance Tuner

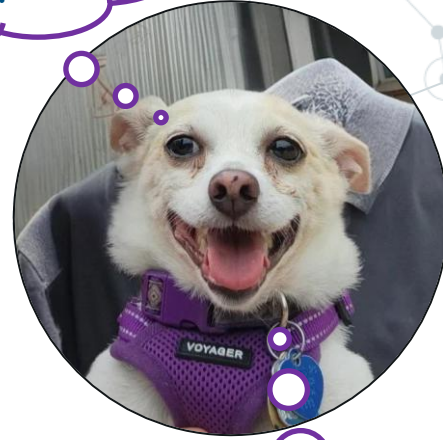
https://youtu.be/lwJe_PZgAN0

Parting Thought

"You don't have to become a storage engineer.

You have to become the person who knows which engineer to go bother."

I know who to go to...



*Deborah!
She's my
favorite human!*

Learn More: Resources

How It Works: Bob Dorr's SQL Server I/O Presentation

<https://techcommunity.microsoft.com/blog/sqlserversupport/how-it-works-bob-dorrs-sql-server-io-presentation/316031>

Troubleshoot slow SQL Server performance caused by I/O issues

<https://learn.microsoft.com/en-us/troubleshoot/sql/database-engine/performance/troubleshoot-sql-io-performance>

How It Works: How is SQL Server Error 833, 15 Sec I/O Detected

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Understanding Network Latency and Impact on Availability Group Replication

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Learn More: Resources

The Ultimate Guide to Process Monitor

<https://adamtheautomator.com/procmon/>

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https://virtualgeek.typepad.com/virtual_geek/2009/06/vmware-io-queues-micro-bursting-and-multipathing.html

Storport Queue Management

<https://learn.microsoft.com/en-us/windows-hardware/drivers/storage/storport-queue-management>

See you
again soon!

Thank You!

Any Questions?



Keep in Touch:

SQL Slack Community

<https://dbatools.io/slack/>

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